28.1: Most eukaryotes are single-celled organisms

Protists—along with plants, animals, and fungi—are eukaryotes; they are in domain Eukarya, one of the three domains of life. Genetic and morphological studies have shown that some protists are more closely related to plants, fungi, or animals than they are to other protists. As a result, the kingdom in which all protists once were classified, Protista, has been abandoned, and various protist lineages are now recognized as major groups in their own right. Most biologists still use the term *protist*, but only as a convenient way to refer to eukaryotes that are not plants, animals, or fungi.

Unlike the cells of prokaryotes, the cells of protists and other eukaryotes have a nucleus and other membrane-enclosed organelles, such as mitochondria and the Golgi apparatus. Such organelles provide specific locations where particular cellular functions are accomplished, making the structure and organization of eukaryotic cells more complex than those of prokaryotic cells. Eukaryotic cells also have a well-developed cytoskeleton that extends throughout the cell (see Figure 6.20). The cytoskeleton provides the structural support that enables eukaryotic cells to have asymmetric (irregular) forms, as well as to change in shape as they feed, move, or grow. In contrast, prokaryotic cells lack a well-developed cytoskeleton, thus limiting the extent to which they can maintain asymmetric forms or change shape over time.

We'll survey the diversity of eukaryotes throughout the rest of this unit, beginning in this chapter with the protists. As you explore this material, bear in mind that

- the organisms in most eukaryotic lineages are protists, and
- most protists are unicellular.

Thus, life differs greatly from how most of us commonly think of it. The large, multicellular organisms that we know best (plants, animals, and fungi) are the tips of just a few branches on the great tree of life (see Figure 26.21).

HHMI BioInteractive Video

Seeing the Invisible: Van Leeuwenhoek's First Glimpses of the Microbial World

image

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Structural and Functional Diversity in Protists

Given that they are classified in a number of different groups, it isn't surprising that few general characteristics of protists can be cited without exceptions. In fact, protists exhibit more structural and functional diversity than the eukaryotes with which we are most familiar—plants, animals, and fungi.

For example, most protists are unicellular, although there are some colonial and multicellular species. Single-celled protists are justifiably considered the simplest eukaryotes, but at the cellular level, many protists are very complex—the most elaborate of all cells. In multicellular organisms, essential biological functions are carried out by organs. Unicellular protists carry out the same essential functions, but they do so using subcellular organelles, not multicellular organs. The organelles that protists use are mostly those discussed in Figure 6.8, including the nucleus, endoplasmic reticulum, Golgi apparatus, and lysosomes. Certain protists also rely on organelles not found in most other eukaryotic cells, such as contractile vacuoles that pump excess water from the protistan cell (see Figure 7.13). One group of dinoflagellate protists has the most complex subcellular structure known: a unique, eye-like organelle called an *ocelloid* with components resembling the lens and retina in the eyes of vertebrates and some other animals **(Figure 28.2)**.

Figure 28.2

The eye-like ocelloid of *Erythropsidinium*, a dinoflagellate.

A micrograph shows the eye like ocelloid of Erythropsidinium, a dinoflagellate, depicting complex protist cellular structure.

This complex, subcellular structure resembles the camera-like eye found in vertebrates and some molluscs. In some dinoflagellates, the ocelloid may function in prey detection.

A micrograph shows a subcellular structure resembling a lateral view of an eye-like structure, termed ocelloid. It has an irregular outline with distinct layers from right to left. The first layer is followed by an elliptical structure labeled the lens, followed by circular and curved layers. A thick, curved elongated structure is present labeled the retina and seems to be embedded in underlying subcellular structures.

Protists are also very diverse in their nutrition. Some protists are photoautotrophs and contain chloroplasts. Some are heterotrophs, absorbing organic molecules or ingesting larger food particles. Still other protists, called **mixotrophs**, combine photosynthesis and heterotrophic nutrition. Photoautotrophy, heterotrophy, and mixotrophy have all arisen independently in many different protist lineages.

Reproduction and life cycles also are highly varied among protists. Some protists are only known to reproduce asexually; others can also reproduce sexually or at least employ the sexual processes of meiosis and fertilization. All three basic types of sexual life cycles (see Figure 13.6) are represented among protists,

along with some variations that do not quite fit any of these types. We will examine the life cycles of several protist groups later in this chapter.

Endosymbiosis in Eukaryotic Evolution

What gave rise to the enormous diversity of protists that exist today? There is abundant evidence that much of protistan diversity has its origins in **endosymbiosis**, a relationship between two species in which one organism lives inside the cell or cells of another organism (the host). In particular, as we discussed in Concept 25.3, structural, biochemical, and DNA sequence data indicate that mitochondria and plastids are derived from bacteria that were engulfed by the ancestors of early eukaryotic cells. The evidence also suggests that mitochondria evolved before plastids. Thus, a defining moment in the origin of eukaryotes occurred when a host cell engulfed a bacterium that would later become an organelle found in all eukaryotes —the mitochondrion.

To determine which bacterial lineage gave rise to mitochondria, researchers have compared the DNA sequences of mitochondrial genes (mtDNA) to those found in major clades of bacteria. In the **Scientific Skills Exercise**, you will interpret one such set of DNA sequence comparisons. Collectively, such studies indicate that mitochondria arose from an alpha proteobacterium (see Figure 27.17). Results from mtDNA sequence analyses also indicate that the mitochondria of protists, animals, fungi, and plants descended from a single common ancestor, thus suggesting that mitochondria arose only once over the course of evolution. Similar analyses provide evidence that plastids descended from a single common ancestor—a cyanobacterium that was engulfed by a eukaryotic host cell.

Scientific Skills Exercise

Interpreting Comparisons of Genetic Sequences

AP Science Practices 2, 3, 4

Which Bacteria Are Most Closely Related to Mitochondria? Early eukaryotes acquired mitochondria by endosymbiosis: A host cell engulfed an aerobic bacterium that persisted within the cytoplasm to the mutual benefit of both cells. In studying which living bacteria might be most closely related to mitochondria, researchers compared DNA sequences encoding ribosomal RNA (rRNA), a component of ribosomes. Ribosomes perform critical cell functions. Hence, genes encoding rRNA are under strong selection and change slowly over time, making them suitable for comparing even distantly related species. In

this exercise, you will interpret some of the research data to draw conclusions about the phylogeny of mitochondria.

A field of wheat.

Wheat, used as the source of mitochondrial rRNA

How the Research Was Done Researchers isolated and cloned DNA sequences from the gene that codes for the small-subunit rRNA molecule for the wheat (a eukaryote) mitochondrion and for five bacterial species:

- Wheat, used as the source of genes encoding mitochondrial rRNA
- *Agrobacterium tumefaciens*, an alpha proteobacterium that lives within plant tissue and produces tumors in the host
- *Comamonas testosteroni*, a beta proteobacterium
- *Escherichia coli*, a well-studied gamma proteobacterium that inhabits human intestines
- *Mycoplasma capricolum*, a gram-positive mycoplasma, which is the only group of bacteria lacking cell walls
- *Anacystis nidulans*, a cyanobacterium

Data from the Research Gene sequences encoding rRNA for the wheat mitochondrion and the five bacteria were aligned and compared. The data table below, called a *comparison matrix*, summarizes the comparison of 617 nucleotide positions from the gene sequences. Each value in the table is the percentage of the 617 nucleotide positions for which the pair of organisms have the same base. Any positions that were identical across all six rRNA-encoding gene sequences were omitted from this comparison matrix.

A table illustrates the similarity matrix score for wheat mitochondrion, A. tumefaciens, C. testosteroni, E. coli, M. capricolum, and A. nidulans.

	Wheat mitochondrion	**A. tumefaciens**	**C. testosteroni**	**E. coli**	**M. capricolum**	**A. nidulans**
Wheat mitochondrion	-	48	38	35	34	34
A. tumefaciens		-	55	57	52	53
C. testosteroni			-	61	52	52
E. coli				-	48	52

M. capricolum					-	50
A. nidulans						-

Data from D. Yang et al., Mitochondrial origins, *Proceedings of the National Academy of Sciences USA* 82:4443–4447 (1985).

INTERPRET THE DATA

- First, make sure you understand how to read the comparison matrix.
- Find the cell that represents the comparison of *C. testosteroni* and *E. coli*. What value is given in this cell?
- What does that value signify about the comparable gene sequences in those two organisms?
- Explain why some cells have a dash rather than a value.
- Why are some cells shaded gray, with no value?
- Why did the researchers choose one plant mitochondrion and five bacterial species to include in the comparison matrix?
-
    - Which bacterium has an rRNA-encoding gene that is most similar to that of the wheat mitochondrion?
- What is the significance of this similarity?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Progress has also been made toward identifying the host cell that engulfed an alpha proteobacterium, thereby setting the stage for the origin of eukaryotes. For example, researchers recently reported the discovery of a new group of archaea, the lokiarchaeotes. In phylogenomic analyses, this group was identified as the sister group of the eukaryotes and its genome was found to encode many eukaryote-specific features. Was the host cell that engulfed an alpha proteobacterium a lokiarchaeote? While this could be correct, it is also possible that the host belonged to a different lineage of archaea or was closely related to the archaea (but was not itself an archaean). In any case, current evidence indicates that the host was a relatively complex cell in which certain features of eukaryotic cells had evolved, such as a cytoskeleton that enabled it to change shape (and thereby engulf the alpha proteobacterium).

Plastid Evolution: A Closer Look

As you've seen, current evidence indicates that mitochondria are descended from a bacterium that was engulfed by a host cell that was an archaean (or a close relative of the archaea). This event gave rise to the eukaryotes. There is also much evidence that later in eukaryotic history, a lineage of heterotrophic eukaryotes acquired an additional endosymbiont—a photosynthetic cyanobacterium—that then evolved into plastids. According to the hypothesis illustrated in **Figure 28.3**, this plastid-bearing lineage gave rise to two lineages of photosynthetic protists, or ********algae**: red algae and green algae.

Figure 28.3

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Diversity of plastids produced by endosymbiosis.
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Through different pathways, primary and secondary endosymbiosis result in stramenopiles, alveolates, euglenids and chlorarachniophytes.

Studies of plastid-bearing eukaryotes suggest that plastids evolved from a cyanobacterium that was engulfed by an ancestral heterotrophic eukaryote (primary endosymbiosis). That ancestor then diversified into red algae and green algae, some of which were subsequently engulfed by other eukaryotes (secondary endosymbiosis).

Heterotrophic eukaryote (large cell with nucleus) engulfs a cyanobacterium (much smaller cell with no nucleus) in primary endosymbiosis. The cyanobacterium inside the eukaryote has three membranes (labeled 1, 2 and 3); one of which will be lost in red and green algal descendants. Two pathways follow:

(1) Cell goes to a red alga, which undergoes secondary endosymbiosis, and is engulfed by another large eukaryotic cell. That cell goes to (a) Stramenopiles with plastid and nucleus and (b) Alveolates with plastid, nucleus and flagellum.

(2) Cell goes to a green alga, and two additional pathways diverge. (2a) Green alga undergoes secondary endosymbiosis and is engulfed by a larger eukaryotic cell, resulting in the Euglenids with plastid, nucleus and flagella. (2b) Green alga undergoes secondary endosymbiosis and is engulfed by a larger eukaryotic cell, resulting in Chlorarachniophytes with plastid and nucleus and pseudopodia.

VISUAL SKILLS *Based on this diagram, which of the following groups are likely to be more closely related: stramenopiles and alveolates, or euglenids and chlorarachniophytes? Explain.*

*For suggested answer, see**Appendix A*.

Let's examine some of the steps in Figure 28.3 more closely. First, recall that cyanobacteria are gram-negative and that gram-negative bacteria have two cell membranes, an inner plasma membrane and an outer membrane that is part of the cell wall (see Figure 27.3). Plastids in red algae and green algae are also

surrounded by two membranes. Transport proteins in these membranes are homologous to proteins in the inner and outer membranes of cyanobacteria, providing further support for the hypothesis that plastids originated from a cyanobacterial endosymbiont.

On several occasions during eukaryotic evolution, red algae and green algae underwent **secondary endosymbiosis**, meaning they were ingested in the food vacuoles of heterotrophic eukaryotes and became endosymbionts themselves. For example, protists known as chlorarachniophytes likely evolved when a heterotrophic eukaryote engulfed a green alga. Evidence for this process can be found within the engulfed cell, which contains a tiny vestigial nucleus, called a *nucleomorph***(Figure 28.4)**. Genes from the nucleomorph are still transcribed, and their DNA sequences indicate that the engulfed cell was a green alga.

Figure 28.4

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Nucleomorph within a plastid of a chlorarachniophyte.
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Image of a nucleomorph within a plastid of a chlorarachniophyte. Nucleomorph is surrounded by inner and outer plastid membrane and nuclear pore-like gap is present.

Four Supergroups of Eukaryotes

Our understanding of the evolutionary history of eukaryotic diversity has been in flux in recent years. Not only has kingdom Protista been abandoned, but other hypotheses have been discarded as well. For example, many biologists once thought that the first lineage to have diverged from all other eukaryotes was the *amitochondriate protists*, organisms without conventional mitochondria and with fewer membrane-enclosed organelles than other protist groups. But recent structural and DNA data have undermined this hypothesis. Many of the so-called amitochondriate protists have been shown to have mitochondria—though reduced ones—and some of these organisms are now classified in distantly related groups.

The ongoing changes in our understanding of the phylogeny of protists pose challenges to students and instructors alike. Hypotheses about these relationships are a focus of scientific activity, changing rapidly as new data cause previous ideas to be modified or discarded. We'll focus here on one current hypothesis: the four supergroups of eukaryotes shown in **Figure 28.5**. Because the root of the eukaryotic tree is not known, all four supergroups are shown as diverging simultaneously from a common ancestor. We know that this is not correct, but we do not know which supergroup was the first to diverge from the other three. In addition, major new groups of protists have been discovered in recent years, but their relationship to the supergroups shown in Figure 28.5 has yet to be resolved. Examples of such unresolved groups include haptophytes and cryptophytes **(Figure 28.6)**, as well as the hemimastigophores, a novel group described in 2018. Overall, as

you read this chapter, it may be helpful to focus less on the specific names of groups of organisms and more on why the organisms are important and how ongoing research is elucidating their evolutionary relationships.

Figure 28.5

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Exploring Protistan Diversity
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A phylogenetic tree of protistan diversity. Examples and descriptions for three of the supergroups.

The following tree represents a phylogenetic hypothesis for the relationships among eukaryotes on Earth today. The eukaryotic groups at the branch tips are related in larger “supergroups,” labeled vertically at the far right of the tree. Groups that were formerly classified in the kingdom Protista are highlighted in yellow. Dotted lines indicate evolutionary relationships that are uncertain. For clarity, this tree only includes representative clades from each supergroup. In addition, the recent discoveries of many new groups of eukaryotes indicate that eukaryotic diversity is actually much greater than shown here. The diagram is as follows.

Branch point 1

Super Group Excavata (Branch point 2 a)

Branch point 3 a

Diplomonads (Protista)

Parabasalids (Protista)

Euglenozoans (Protista)

Super Group “SAR” clade (Branch point 2 b, dotted line)

Branch point 3 b 1 (dotted line)

(Stramenopiles dotted line) Branch point 4 b 1

Diatoms (Protista)

Golden algae (Protista)

Brown algae (Protista)

(Alveolates, dotted line) Branch point 4 b 2

Branch point 5 b 1

Dinoflagellates (Protista)

Apicomplexans (Protista)

Ciliates (Protista)

(Rhizarians, dotted line) Branch point 3 b 2

Branch point 4 b 3

Radiolarians (Protista)

Forams (Protista)

Cercozoans (Protista)

Super Group Archaeplastida (Branch point 2 c)

Branch point 3 c

(Green algae) Chlorophytes (Protista)

Branch point 4c

(Green algae) Charophytes (Protista)

Plants

Red algae (Protista)

Super Group Unikonta (Branch point 2 d, dotted line)

(Amoebozoans, dotted line) Branch point 3 d 1

Slime molds (Protista)

Tubulinids (Protista)

Entamoebas (Protista)

(Opisthokonts, dotted line) Branch point 3 d 2

Branch point 4 d 1

Nucleariids (Protista)

Fungi

Branch point 4 d 2

Choanoflagellates (Protista)

Animals

Notes on Excavata are as follows. A micrograph shows Giardia, which is roughly tear-shaped single cell with several attached string-like structures (flagella). Scale: 5 micrometers. Excavata: Some members of this supergroup have an “excavated” groove on one side of the cell body. Two major clades (the parabasalids and diplomonads) have highly reduced mitochondria; members of a third clade (the euglenozoans) have flagella that differ in structure from those of other organisms. Excavates include parasites such as Giardia, as well as many predatory and photosynthetic species. Photo: Giardia intestinalis, a diplomonad parasite. This diplomonad (colorized SEM), which lacks the characteristic surface groove of the Excavata, inhabits the intestines of mammals. It can infect people when they drink water contaminated with feces containing Giardia cysts. Drinking such water, even from a seemingly pristine stream, can cause severe diarrhea. Boiling the water kills the parasite.

Parts are as follows.

SAR clade. Image: Diatom diversity and Globigerina, a rhizarian in the SAR clade. This supergroup contains (and is named after) three large and very diverse clades: Stramenopila, Alveolata, and Rhizaria. Stramenopiles include some of the most important photosynthetic organisms on Earth, such as the diatoms shown here. Alveolates also include photosynthetic species, as well as important pathogens, such as Plasmodium, which causes malaria. According to one current hypothesis, stramenopiles and alveolates originated by secondary endosymbiosis when a heterotrophic protist engulfed a red alga. Photo. Diatom diversity. These beautiful single-celled protists are important photosynthetic organisms in aquatic communities (LM). The rhizarian subgroup of SAR includes many species of amoebas, most of which have pseudopodia that are threadlike in shape. Pseudopodia are extensions that can bulge from any portion of the cell; they are used in movement and in the capture of prey. Photo. Globigerina, a rhizarian in SAR. This species is a foram, a group whose members have threadlike pseudopodia that extend through pores in the shell, or test (LM). The inset shows a foram test, which is hardened by calcium carbonate.

Archaeplastida. This supergroup of eukaryotes includes red algae and green algae, along with plants. Red algae and green algae include unicellular species, colonial species, and multicellular species (including the green alga Volvox). Many of the large algae known informally as “seaweeds” are multicellular red or green algae. Protists in Archaeplastida include key photosynthetic species that form the base of the food web in many aquatic communities. Photo. Volvox, a multicellular freshwater green alga. This alga has two types of differentiated cells, and so it is considered multicellular rather than colonial. It resembles a hollow ball whose wall is composed of hundreds of biflagellated cells (see inset LM) embedded in a gelatinous extracellular matrix; if isolated, these cells cannot reproduce. However, the alga also contains cells that are

specialized for either sexual or asexual reproduction. The large algae shown here will eventually release the small “daughter” algae that can be seen within them (LM).

Unikonta. This supergroup of eukaryotes includes amoebas that have lobe- or tube-shaped pseudopodia, as well as animals, fungi, and non-amoeba protists that are closely related to animals or fungi. According to one current hypothesis, the unikonts were the first eukaryotic supergroup to diverge from all other eukaryotes; however, this hypothesis has yet to be widely accepted. Photo. A unikont amoeba. This amoeba, the tubulinid Amoeba proteus, is using its pseudopodia to move.

DRAW IT *Draw a simplified version of this phylogenetic tree that shows only the four supergroups of eukaryotes. Now sketch how the tree would look if the unikonts were the sister group to all other eukaryotes.*

*For suggested answer, see**Appendix A*.

Video

Volvox

image

Play

image

Video

Amoeba Pseudopodium

image

Play

image

Figure 28.6

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A haptophyte and a cryptophyte.
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Part a. A micrograph of a spherical haptophyte. Part b. A micrograph of cryptophytes with flagella.

Although they are major groups of protists, haptophytes and cryptophytes are not shown in Figure 28.5 because their phylogenetic relationships have yet to be resolved. ***(a)*** *The haptophyte Emiliania huxleyi is a*

key producer in the ocean. ***(b)*** *The outer surface of this photosynthetic cryptophyte is covered with protective plates. (SEM)*

Part a. A micrograph of a spherical haptophyte, Emiliania huxleyi. It resembles a globe-shape with many overlapping subunits. Each subunit resembles two circles, attached by radiating lines (similar to a wheel). Scale bar 2 micrometers.

Part b. A micrograph of cryptophytes with flagella. An image of an elliptical organism with a tail-like flagellum is shown. The oval part is covered by scales. Scale bar: 8 micrometer.